Virtual University of Pakistan

(zoo501) (Developmental biology)

Final term Past Papers

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2.What are silensors... 2
3.What is embryonic shield... 2
4.Diffr axoneme n dyenin... 2
5.Role of calcium in sperm n egg fusion... 3
6.What r lineages to generate skeleton... 3
8.What do u know about myogenic BHLHL.. 3
9.Diffrnc b/w identical n fratrnal twinz... 5
Define.. Area opaca.. Area pellicuda.. Sub marginal laye.. Germinal line.... 5
10.osteogenesis.... 10

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(subjective)
Q.1 define silencers?2
Q.2 what is primitive node or henson node?2
Q.3 define periosteun?2
Q.4 difference b/w identical and fraternal twins?3
Q.5 differnce b/w gray matter and white matter?3
Q.6 factors of gastrulation periods?3
Q.7 interaction of sperm and egg steps?5

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Q.8 short note on fertilization?3

Q.9 domains of neural crest?5

Q.10 process of osteogensis or bone formation?10

objective(m.c.q)

1.mode of commitment.....ans.3

2.strageries of commitment.....ans.2

3.insects type of specification.....ans.synctial specification

4.rana pipiens eggg.....ans.2500

5.multinucleated type of cell form.....ans.syncytiotrophoblast

6.slow block polyspermy is .....ans.calcium ion

7.process of osteogensis.....ans.2

8.spiral holoblastic cleavage in.....ans.snail

9.camptomelic dysplasia cause .....ans.sox9 mutation

10.human brain contains.....ans.10^12 glial cells

11.telolecithal is cleavage in.....ans.fishes

1: What is blastocoel?

2: Write the general consept of fertilization?

3: What is marginal belt?

4:Write three distinct lineage of Osteogenesis?

5: What is syncytiotrophoblast and its function?

6:Write the function of Metencephalon?

7:Write the role of calcium in the fusion of egg and sperm?

8:Describe the process of gastrulation? is Question ki exact staement yad ni

9:Enlist the 4 factors which cause the movement in gene regulation? Is question exact statement

1. difference b/w primitive node & Henson node? 3

2. process of gastrulation?2

3.Differ b/w evolutionary embryology & tetralogy?2

4. Domains of transcription factor?5

5. Domains of neural crest?5

6. Mitosis promoting factor?3

7. Factors of neural tube?5

8. Domains of neural crest?5

Mcqs

1. Fg7,TGF5……..

2.about leopard male frog……

3. embryo layers……
4.Gastrulation cell stages……..
5. Zp3 function in mouse……
6. sea urchin eggs……

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Neural crest =2

Development k bry mn 2 nb ka q tha yd ni

General concepts of fertilization =3

Neural ring =2

Domian k bary mn 3 mrk ka q tha but yd ni ki ki domain?

Marble bone disease =3

Syncytiotrophoblast =5

Function of domain of neural crest =5

Factors which cause movement of gestrulation period =5

Detail note on osteogenesis =10

MCQ'S

1Pax6, pax10 or sox10 k bary ik ik mcq tha..

1Koi sciecntific nam dy ka animal k eggs ki tadad pochi thi yd ni

3Even distribution of yolk in ovum ans:isolecithl

4Yolk is present in centrolacithl ans:center

5Egg acitvates due to which ion more option: ca,K,Na,Mg

6Which drm give rise to brain

7Which drm give rise to eye

8How mny methods of commitment are ans:2

9Mod of commitment in insects ans:syncytal specification

10Regulative developmnt arises from ans:conditional specification

11,12Domain, bird k bary ik ik mcq tha

Nuclus hn DNA k bps k tadad pr mcq tha 13when K ions leave sperm membrane poteitial is ?

1) what is naive? 2 marks

2) Define Medial hinge point (MHP)? 3 mraks

3) Diff b/w late and early response of egg to sperm. 3 marks

4) General concept of fertilization 3 marks

5) 3 lineages that generate skeleton 3 marks

6)basic modes of commitment 5 marks

7) Dife area opaca,area pellucida,sub-germinal cavity, marginal zone 5 marks

8) functional domains of neural crest 5 marks

9)osteogenesis 10 marks

10) diff b/w identical and fraternal twins 3 marks

Final term 2018

1.Difference between vegetative pole &animal pole

2.karyokinesis & cytokinesis

3.capacitation

4.what happens when egg and sperm meet

5.myogenesis

6.primary neurulation & secondary neurulation

7.functions of zona pellucida

8.neural crust (10 marks)

Write one example of cytoplasmic determiant ? Write strategies of commitment? What is fast block of polyspermy?Differentiate between karyokinesis and cytokinesis? Differentiate between identical and fraternal twins? Write major pathways of differentiation of neural crest? Aik cartilage activating factor wala tha?

2marks

function of vitalline membrane which type of
eggs have meroblastic cleavage

diff b/w cortices and nuclei

3marks

what will happen if blastomere is removed ....in conditional specification

what will happen when egg and sperm fuses name
3 primary vesicle forms from brain

5marks

4 levels of gene expression

which factor control the rate of cell division and position of blastomere in cleavage diff b/w identical and fraternal twins

10 marks

names of patterns of cleavage

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